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| EXP NO: 5 | CLUSTERING WITH K-MEANS AND DIMENSIONALITY REDUCTION WITH PCA |
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### AIM:

To demonstrate the application of Unsupervised Learning models, specifically K-Means clustering for grouping data points and Principal Component Analysis (PCA) for dimensionality reduction and visualization, using a suitable dataset.

### ALGORITHM:

#### 1. K-Means Clustering

K-Means is an iterative clustering algorithm that aims to partition  $n$  observations into  $k$  clusters, where each observation belongs to the cluster with the nearest mean (centroid).

#### Steps:

- 1. Initialization:** Choose  $k$  initial centroids randomly from the dataset.
- 2. Assignment:** Assign each data point to the cluster whose centroid is closest (e.g., using Euclidean distance).
- 3. Update:** Recalculate the centroids as the mean of all data points assigned to that cluster.
- 4. Iteration:** Repeat steps 2 and 3 until the centroids no longer move significantly or a maximum number of iterations is reached.

#### 2. Principal Component Analysis (PCA)

PCA is a statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components.

#### Steps:

- 1. Standardization:** Standardize the dataset (mean = 0, variance = 1).
- 2. Covariance Matrix Calculation:** Compute the covariance matrix of the standardized data.
- 3. Eigenvalue Decomposition:** Calculate the eigenvalues and eigenvectors of the covariance matrix.
- 4. Feature Vector Creation:** Sort the eigenvectors by decreasing eigenvalues and select the top  $k$  eigenvectors to form a feature vector (projection matrix).
- 5. Projection:** Project the original data onto the new feature space using the feature vector.

## CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import make_blobs
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score

print("--- Part 1: K-Means Clustering ---")

X, y = make_blobs(n_samples=300, centers=3, cluster_std=0.60, random_state=42)
df_kmeans = pd.DataFrame(X, columns=['Feature_1', 'Feature_2'])
print("\nOriginal K-Means Dataset Head:")
print(df_kmeans.head())

wcss = []
for i in range(1, 11):

    kmeans = KMeans(n_clusters=i, init='k-means++', max_iter=300, n_init=10,
random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), wcss, marker='o', linestyle='--')
plt.title('Elbow Method for Optimal K (K-Means)')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('WCSS')
plt.grid(True)
plt.show()
```

```

optimal_k = 3
kmeans = KMeans(n_clusters=optimal_k,      init='k-means++', max_iter=300, n_init=10,
random_state=42)
clusters = kmeans.fit_predict(X)
df_kmeans['Cluster'] = clusters


plt.figure(figsize=(10, 8))
sns.scatterplot(x='Feature_1', y='Feature_2', hue='Cluster', data=df_kmeans, palette='viridis',
s=100, alpha=0.8)
plt.scatter(kmeans.cluster_centers_[0], kmeans.cluster_centers_[1], s=300, c='red',
marker='X', label='Centroids')
plt.title(f'K-Means Clustering with K={optimal_k}')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.legend()
plt.grid(True)
plt.show()


silhouette_avg = silhouette_score(X, clusters)
print(f"\nSilhouette Score for K-Means (K={optimal_k}): {silhouette_avg:.3f}")


print("\n--- Part 2: Dimensionality Reduction with PCA ---")

X_pca, y_pca = make_blobs(n_samples=500, n_features=4, centers=4, cluster_std=1.0,
random_state=25)
df_pca_original

df_pca_original = pd.DataFrame(X_pca, columns=[f'Feature_{i+1}' for i in
range(X_pca.shape[1])])
df_pca_original['True_Cluster'] = y_pca
print("\nOriginal PCA Dataset Head:")
print(df_pca_original.head())
print(f"Original PCA Dataset Shape: {df_pca_original.shape}")


scaler = StandardScaler()
X_pca_scaled = scaler.fit_transform(X_pca)


pca = PCA(n_components=2)
principal_components = pca.fit_transform(X_pca_scaled)


df_principal_components = pd.DataFrame(principal_components,
columns=['Principal_Component_1', 'Principal_Component_2'])

```

```

df_principal_components['True_Cluster'] = y_pca
explained_variance = pca.explained_variance_ratio_
print("\nPrincipal Components Head:")
print(df_principal_components.head())
print(f"\nExplained Variance Ratio: {explained_variance}")
print(f"Total Explained Variance by 2 PCs: {explained_variance.sum():.3f}")

plt.figure(figsize=(10, 8))
sns.scatterplot(x='Principal_Component_1', y='Principal_Component_2', hue='True_Cluster',

                data=df_principal_components, palette='Paired', s=100, alpha=0.8)
plt.title('PCA - Dimensionality Reduction to 2 Components')
plt.xlabel(f'PC1 ({explained_variance[0]*100:.2f}%)')
plt.ylabel(f'PC2 ({explained_variance[1]*100:.2f}%)')
plt.grid(True)
plt.show()

kmeans_pca = KMeans(n_clusters=4, init='k-means++', max_iter=300, n_init=10,
random_state=42)
clusters_pca = kmeans_pca.fit_predict(principal_components)
df_principal_components['KMeans_Cluster_on_PCA'] = clusters_pca

plt.figure(figsize=(10, 8))
sns.scatterplot(x='Principal_Component_1', y='Principal_Component_2',
hue='KMeans_Cluster_on_PCA',

                data=df_principal_components, palette='viridis', s=100, alpha=0.8)
plt.scatter(kmeans_pca.cluster_centers_[0], kmeans_pca.cluster_centers_[1], s=300,
c='red', marker='X', label='Centroids')
plt.title('K-Means Clustering on PCA-Reduced Data')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend()
plt.grid(True)
plt.show()

silhouette_avg_pca = silhouette_score(principal_components, clusters_pca)

print(f"\nSilhouette Score for K-Means on PCA-Reduced Data (K=4):
{silhouette_avg_pca:.3f}")

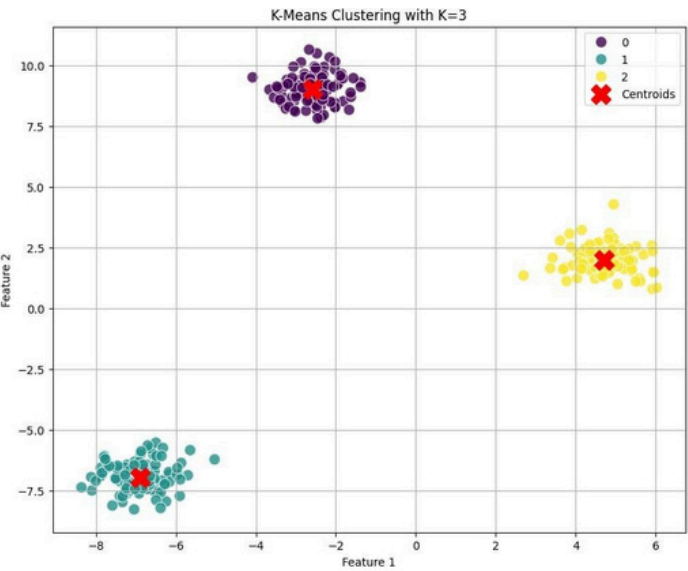
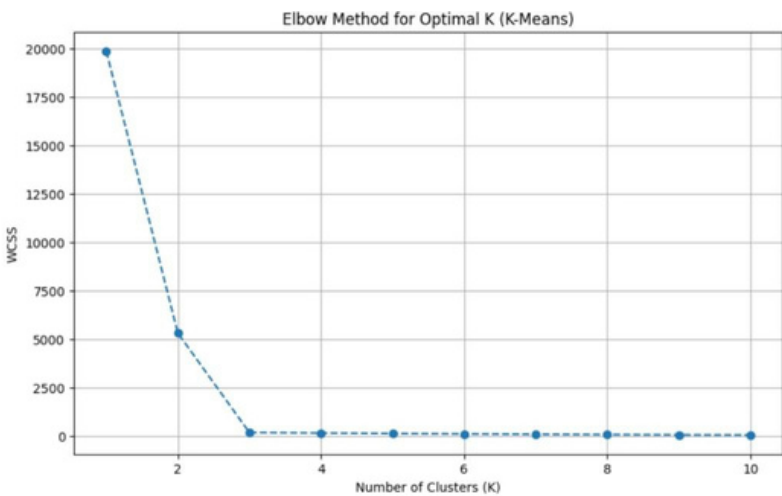
```

OUTPUT:

--- Part 1: K-Means Clustering ---

Original K-Means Dataset Head:

|   | Feature_1 | Feature_2 |
|---|-----------|-----------|
| 0 | -7.155244 | -7.390016 |
| 1 | -7.395875 | -7.110843 |
| 2 | -2.015671 | 8.281780  |
| 3 | 4.509270  | 2.632436  |
| 4 | -8.102502 | -7.484961 |



Silhouette Score for K-Means (K=3): 0.908

--- Part 2: Dimensionality Reduction with PCA ---

Original PCA Dataset Head:

|   | Feature_1 | Feature_2 | Feature_3 | Feature_4 | True_Cluster |
|---|-----------|-----------|-----------|-----------|--------------|
| 0 | -0.638667 | 1.110057  | -6.400722 | -0.204990 | 3            |
| 1 | -2.951556 | -7.657445 | 3.844794  | 0.903589  | 1            |
| 2 | -0.253177 | 2.125103  | -7.869801 | 0.559678  | 3            |
| 3 | -2.151209 | 3.401400  | -5.734930 | 0.965230  | 3            |
| 4 | -2.347519 | -7.230467 | 3.478891  | -0.443440 | 1            |

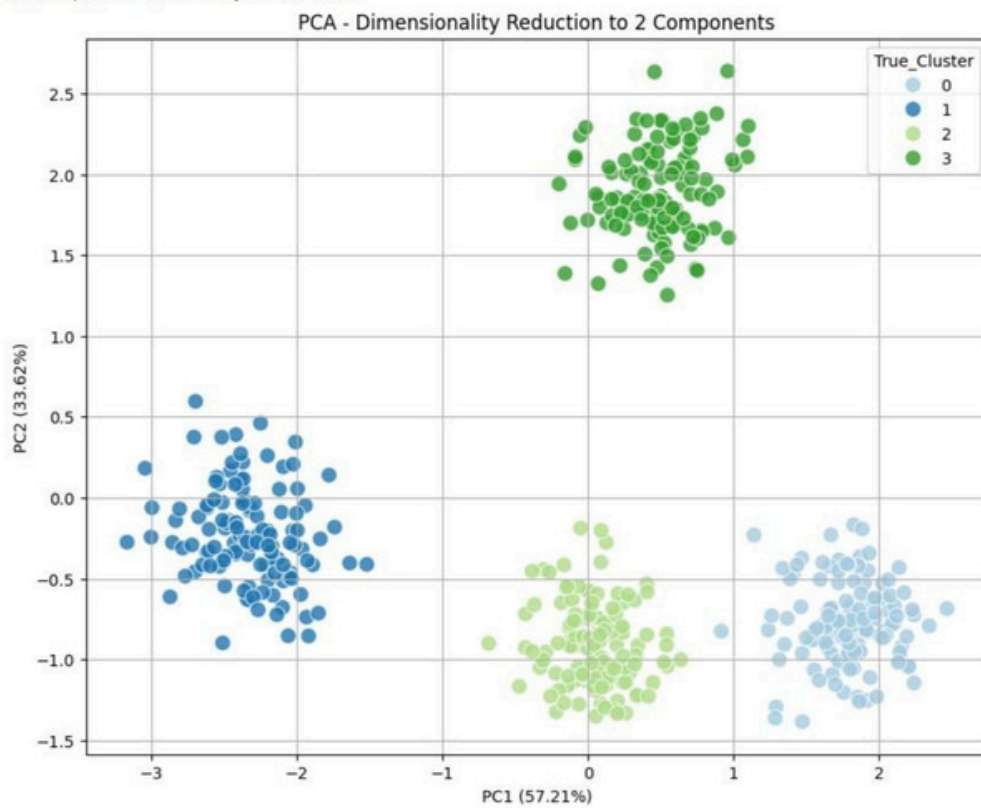
Original PCA Dataset Shape: (500, 5)

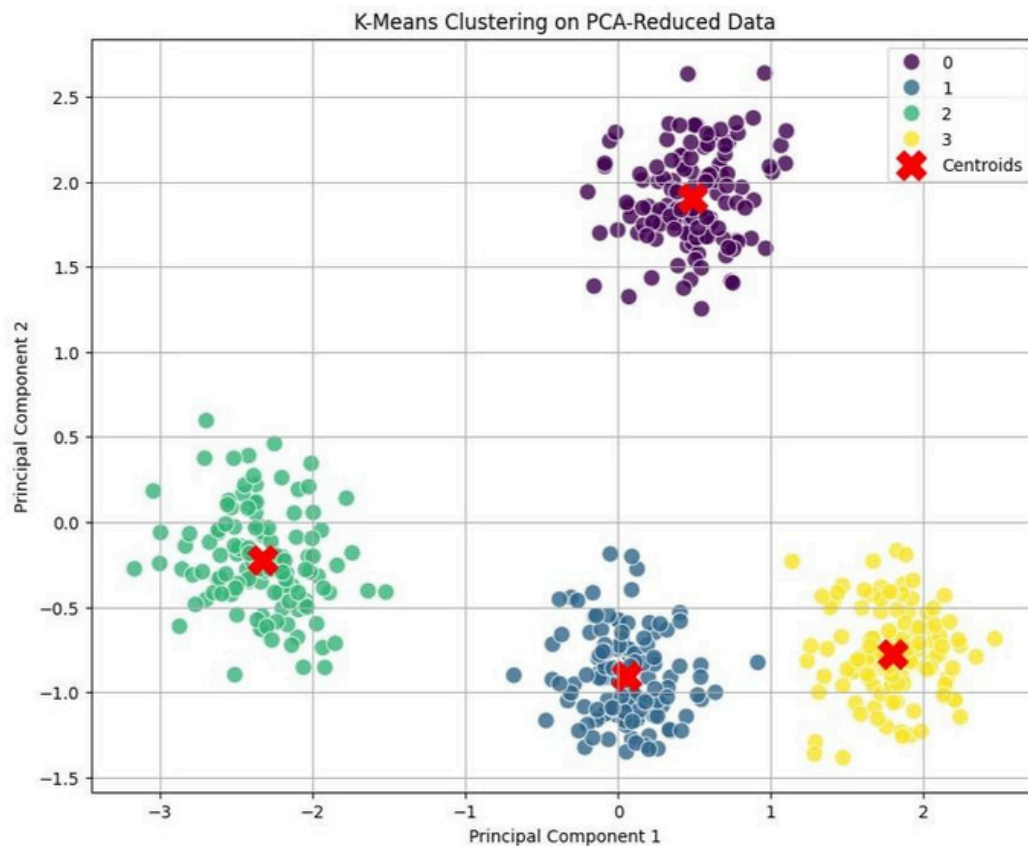
Principal Components Head:

|   | Principal_Component_1 | Principal_Component_2 | True_Cluster |
|---|-----------------------|-----------------------|--------------|
| 0 | 0.455305              | 1.623917              | 3            |
| 1 | -2.705622             | 0.375012              | 1            |
| 2 | 0.810234              | 1.966926              | 3            |
| 3 | 0.427139              | 2.149626              | 3            |
| 4 | -2.407508             | 0.099250              | 1            |

Explained Variance Ratio: [0.57208431 0.33622342]

Total Explained Variance by 2 PCs: 0.908





Silhouette Score for K-Means on PCA-Reduced Data (K=4): 0.776

## RESULT:

The K-Means clustering and Principal Component Analysis (PCA) techniques were successfully implemented on the given dataset.

- **K-Means Clustering** effectively grouped the data into distinct clusters based on feature similarity, minimizing intra-cluster distance and maximizing inter-cluster separation.

**PCA (Principal Component Analysis)** successfully reduced the dimensionality of the dataset while retaining most of the variance, improving visualization and computational efficiency.

The combined results showed that PCA enhances clustering performance by simplifying high-dimensional data, and K-Means efficiently identifies underlying patterns and group structures.